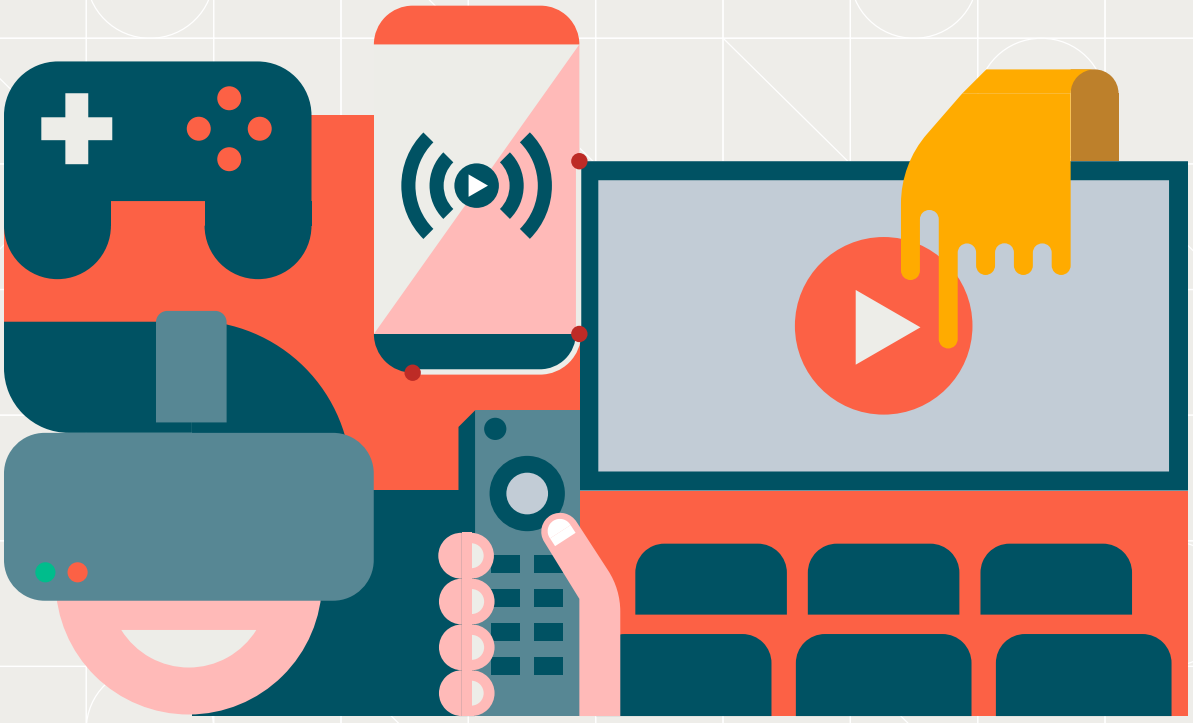
databricks

Whitepaper

Designing a Modern Data Strategy for Premium+ Games

Strategies for engaging and monetizing players after acquisition



databricks

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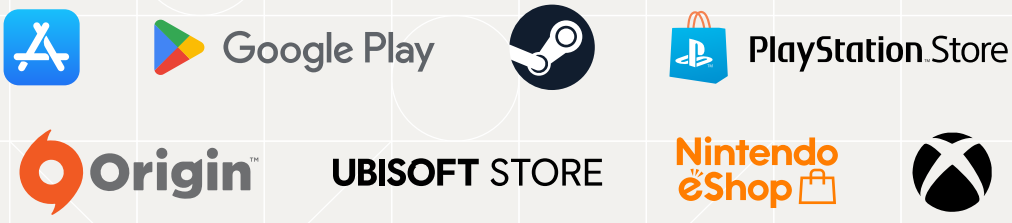
01 Introduction

In the evolving games industry, data analytics is crucial for success. As companies shift from box sales to Live Operations (LiveOps) titles, they face increased data volume and velocity. Leaders use data to boost player engagement, prevent churn and tailor experiences. Because players typically engage with only a few titles, it's vital to spot trends early and personalize interactions. This whitepaper offers strategies to build a robust data approach, enhancing decision-making and player engagement in a competitive landscape.

This whitepaper is geared toward Premium game makers transitioning to a Premium+, F2P, LiveOps model, though mobile games companies will find it familiar. We'll cover strategies for engaging and monetizing players after acquisition.

In this whitepaper we define Premium, Premium+ and Live Operations as follows:

Premium	Premium+	Live Operations (LiveOps)
A game where the player must purchase the title up front to play	A game that has an up-front cost but also has microtransactions	A title that has updates over time to ensure ongoing engagement with the title, typically with the intention of selling microtransactions



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A LiveOps-driven data strategy

LiveOps shifts the focus to convincing a gamer to continuously spend on a title rather than making a singular purchase. This requires a shift for all lines of business at a Premium game studio, including product, pricing, marketing and creative. LiveOps is also different because monetization decisions have to be considered early in development and can't just be added on at the end.

A LiveOps-driven data strategy:

- Puts the continued experience of a player front and center
- Values near real-time insight
- Enables learning from experimentation within an experience
- Enables configuration-driven development
- Can track cohorts through their lifetime
- Is trusted by the organization to shape the future direction of the title

Enhancing player engagement and experience

A deeply engaging and satisfying player experience is key to the success of any game. To create that experience you need actionable insights that are iterated upon based on trusted datasets. Through data you can understand the nuances of player interactions, identify points of friction, understand feedback at scale and uncover opportunities to grow engagement with your players. By analyzing patterns found in session lengths, playstyle, FTUE (first-time user experience) and level completion rates, in-game purchase behaviors and social interactions, developers can fine-tune game mechanics and creative pipeline and reengagement strategies to personalize the player experience and improve engagement.

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Configuration-driven development

In configuration-driven development you take advantage of a series of data points and variables to determine details of the player experience. This methodology has been leveraged within F2P titles for quite a while, particularly with regard to the offers that are presented to a player, whether or not they see ads and the difficulty of nonranked single-player games. Configuration-driven development is a core method by which you personalize the player experience. It's most easily implemented early in the development process. As you build your title, your development and data teams should be collaborating to answer the question, "What aspects of the game would I likely want to personalize for my players and what details about their experience so far might be useful to make the personalization decision."

Some examples include:

- Based on the ad campaign a player clicked through, you might modify their FTUE
- Adjusting difficulty for a level based on how a player performed previously
- Matching a bot with a player who may be struggling with others in their ELO bracket
- Modifying the art style based on a player's physical region or the time of day

Driving monetization strategies

Monetization is the outcome of engagement over time, a meaningful product and signposting to help a player know how, when and why to spend money within the game. A robust monetization strategy is core to any Live Operations title, and data analytics helps optimize revenue-generating opportunities for your strategic objectives. As you consider analytics related to your monetization strategy, think about:

- What's happened in the past
- How to track the impact of change
- How to predict an outcome that inspires a player to take action within the title

The past provides insight into where you might find useful information. Identifying a player with high player lifetime value (pLTV) may inform a decision to provide a personalized VIP experience, but more importantly, this provides insight into what the journey of a high LTV player looks like. With these insights you can adjust how you engage with other players to promote their progression toward a high LTV relationship with your game.

For more information about pLTV, see Appendix: Key Metrics to Consider.

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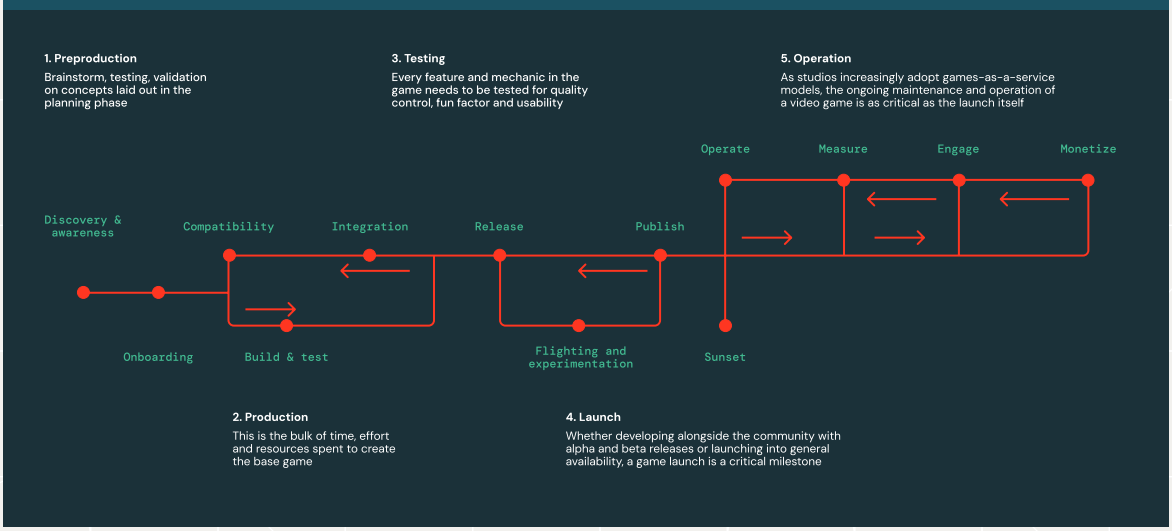
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The iterative nature of Live Operations

Before we dive into the creation of your data strategy, we should further explore one more consideration: the iterative nature of Live Operations. As you progress your title from soft launch to global launch to stabilization, you'll further align your title to your player's preferences. In order to accomplish this you need to do the following:

1. Collect experiential data
2. Generate actionable insight
3. Experiment and modify your title as a result of that insight

Game development lifecycle



Your experiential data will be both qualitative and quantitative and will come in a variety of formats, including tabular, NoSQL, logs, telemetry, surveys and free-form text. See Appendix: Key Metrics to Consider for insight into typical datasets used by games companies. As you collect these datasets you'll explore them through four lenses:

- The current state of your KPIs — how is your game performing
- Potential leading indicators enabling outcome prediction and proactive action
- Harbingers of change that can help you predict things like churn
- Anomaly detection to find fraud, cheaters, VIPs and new growth vectors

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No single journey fits all organizations. This whitepaper brings together best practices that we've seen to help you on your journey, and it's intended to give you pause and help you be thoughtful as you build out your data strategy. While we'll highlight some KPIs, some of which are ubiquitous, we want you to focus on the *why* behind the metric more than the metric itself. Your data and AI strategy will enable you to differentiate and innovate in ways that align with your business goals.

A written data strategy is a valuable tool that helps stakeholders understand key performance areas. It literally defines how metrics are derived and helps build trust within an organization. A strategy without backing and execution will fail. In addition to the work described in this document, consider how you'll implement these systems and your vision, and how you'll build and reinforce a data-informed culture across the organization. Executive alignment, data guilds, data show-tell sessions, newsletters and other communications that share the impact this strategy has had on different lines of business are a few of the ways you can further the success of your strategy and your organization.

Step 1: Build a solid foundation

Outside of revenue makeup, you'll notice that little of this step involves data or metrics. Data doesn't define your direction, it's there to help inform, implement, validate and course-correct your business goals.

DEFINE ORGANIZATIONAL GOALS

Begin by defining your organization's goals for its Premium+ business. Your data strategy will need to align with these goals. Your data strategy is there to help inform, implement, validate and course-correct those business goals.

- How much of your revenue do you expect to come from the Premium+ business in one year, five years and 20 years?
- What are the core tenants you want to ensure as you implement this business structure?
- When considering monetization strategies, what impact will it have on your brand?
- How much backlash are you willing to accept from your superfans as you make the transition to Premium+?
- How can you smooth that transition?

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ORGANIZATION AND CULTURE

Consider how you'll promote data sharing by studios across the organization. By having the organization's datasets available with democratized, governed access, you'll be able to share this benefit with participating studios.

- How will you show your different stakeholders the value of investing in your data platform, strategy and related human resources, and how will you ensure trust in the data and derived insight?
- How willing are your studios to share information about players, their experiences and their interests with each other to help grow the organization as a whole and to ensure the best experience for your players?
- If you have a contentious organization, what can the central team do to bridge that gap?
- How can you bring these teams together and show them the value to the player of working together?
- If they're unwilling or unable to share raw data, could your studios share insight about players across organizations?

IDENTITY RESOLUTION

How will you create a 360-degree view of the player across your titles? A common approach is an organization-level ID. The challenge is having players opt in. How will you incentivize it? What could you offer in exchange? Any time you're discussing player data, be sure to talk with your legal team.

- How would you roll out a user ID to your players?
- What value exchange could you offer to them for creating the ID?

TECHNICAL EVALUATION

Where is your data today? How will you gain access to it and share it across your organization? Have you outlined the common use cases for your teams and ensured your platform can deliver on them? Answering these questions will help ensure you have the right platform and give teams insight into prevalent use cases across the organization that they may have been unaware of. Additional questions to consider:

- Who produces the data?
- What systems contain it?
- Do you have the team needed to consolidate data or insight from many sources into a place where each of your studios and your central teams can access, investigate and learn from it in a governed and secure manner?

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Step 2: Define your lexicon

Some metrics are the gold standard and serve as measuring sticks across studios and titles to understand business performance across disparate teams. These core metrics need to be described and defined within your data strategy. You'll need to define each metric, so it can be discussed externally, with the formulas and data sources that'll be used to calculate it. To avoid confusion, this level of standardization and explicit definition is critical in your strategy. Take, for example, the KPI pLTV: it could mean player LTV (the current, real, lifetime value of all players) or predicted LTV.

SEEK INTERNAL CONSENSUS

Work with internal business leaders to determine which metrics are most important to your organization. See Appendix: Key Metrics to Consider for example definitions and calculations.

CONSIDER USERS, USAGE AND PEER GROUPS

Identify and engage with a cross-functional set of leaders from Game Dev, Marketing, Revenue and Operations to ensure all teams are served by the systems and metrics that you develop. Consider meeting with your peers and partners like Databricks across the industry to discover questions you might not be asking.

EXAMPLE METRICS

In Appendix: Key Metrics to Consider, we've compiled a series of common engagement and monetization metrics needed for a LiveOps-focused title, including a high-level definition of the metric and why it matters. We separate these metrics into two separate categories: engagement and monetization. Engagement metrics enable the monitoring of users, sessions, the player journey and their experience. Monetization metrics focus on measuring the effectiveness of user acquisition and revenue generation efforts.

Note: Most of these metrics can be viewed through two separate lenses: a user/player (nonpaying) and a customer. Usually you'll track your messages through both lenses. Player metrics will help when considering the cost of operations, your sustainability and conversion rates. Customer metrics will help you understand how engaged your spenders are, how revenue is distributed, where to put investments and where to set your VIP line. Great examples of customer metrics are ARPU (average revenue per user), ARPPU (average revenue per paying user), DAU (daily active users) and DAC (daily active customers).

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ALLOW FOR FLEXIBILITY

To enable comparisons across teams, you'll use a limited set of consistent data points to build standard definitions for KPIs used with your executive teams. A game team will have additional data available to them that could be used to provide a more nuanced view of that KPI. A great example of this is pLTV. The central KPI may only include purchases made and universal milestones, whereas your studio may include session details and title-specific milestones to give them a more detailed view they can use to drive their title's growth. This extra depth is especially useful for identifying high-value players more quickly or addressing concerns with gameplay related to revenue conversion.

Step 3: Establish benchmarks and targets

Performance-related targets and benchmarks vary widely across game genres, platforms, regions, budgets and device types. Ensure that these benchmarks are realistic, attainable and aligned with your studio's realities. Your benchmarks and targets are a way to ensure the profitability of your organization, but when done wrong, they can be a heavy hit to morale.

SEEK INTERNAL CONSENSUS

Work with internal business leaders to determine which metrics are most important to your organization. Leverage resources like the information in Appendix: Key Metrics to Consider of this whitepaper and other industry publications to prompt discussion with your teams.

SEEK EXTERNAL BENCHMARKS

Look at industry benchmarks from sources like Sensor Tower (including data.ai and formerly App Annie), Naavik and Newzoo. Do your best to focus on comparisons most relevant to the game or genre you're building. These benchmarks will help you understand if you're in the right ballpark, but they shouldn't be used too literally because the data can be wrong and/or inferred by the data provider.

TARGET CREATION

The more specific and measurable you make the target, the easier it'll be to analyze performance against it. For games where long development cycles and complex launch schedules are common, it's important to align metrics to timelines and game milestones.

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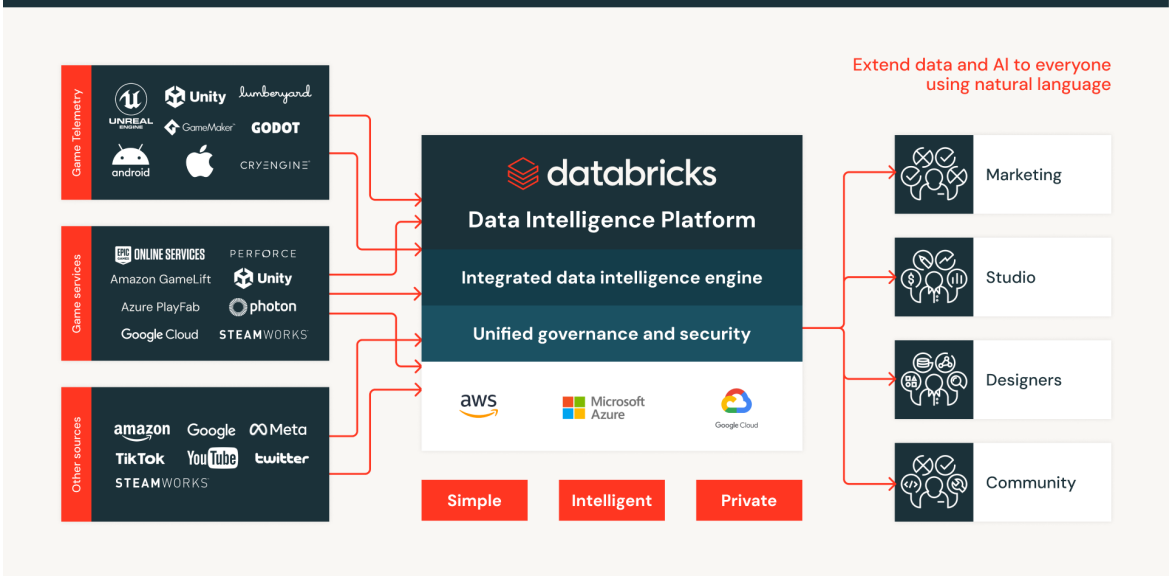
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LEADING INDICATORS

Most metrics provide a view of what's happened in the past. While these are important to understand the current health of your title, they don't show trends you may want to change. Some metrics, like D1 retention, can be leading indicators on their own, but most leading indicators describe the changing vector of a lagging indicator over time. For example, what's the month-to-month change in DAU, level completion rate or D1 retention? When working with leading indicators it's critical to have strong data governance, trust in your data and the ability for users of these metrics to interrogate the underlying math used to calculate these changes. These indicators will be some of the most interesting to your team, so invest time to ensure they can derive full value from them.

Databricks democratizes data and AI



Step 4: Define data sources and collection methods

DATA SOURCES AND QUALITY DEFINITIONS

For your golden metrics, where consistency is important, define both the definition of the metric and the appropriate data sources for its calculation. You'll also want to define the freshness of that metric to ensure that it's trusted by the team. You'll have to consider your organizational structure, especially as it relates to your data organization.

- Do studios have their own data team?
- Do they have their own data engineers or just data analysts?
- Who will support them at HQ to ensure that this data stream is connected correctly?
- Who do they contact when something goes wrong?
- How will you validate that the data sources are correct and that you're receiving the right information?

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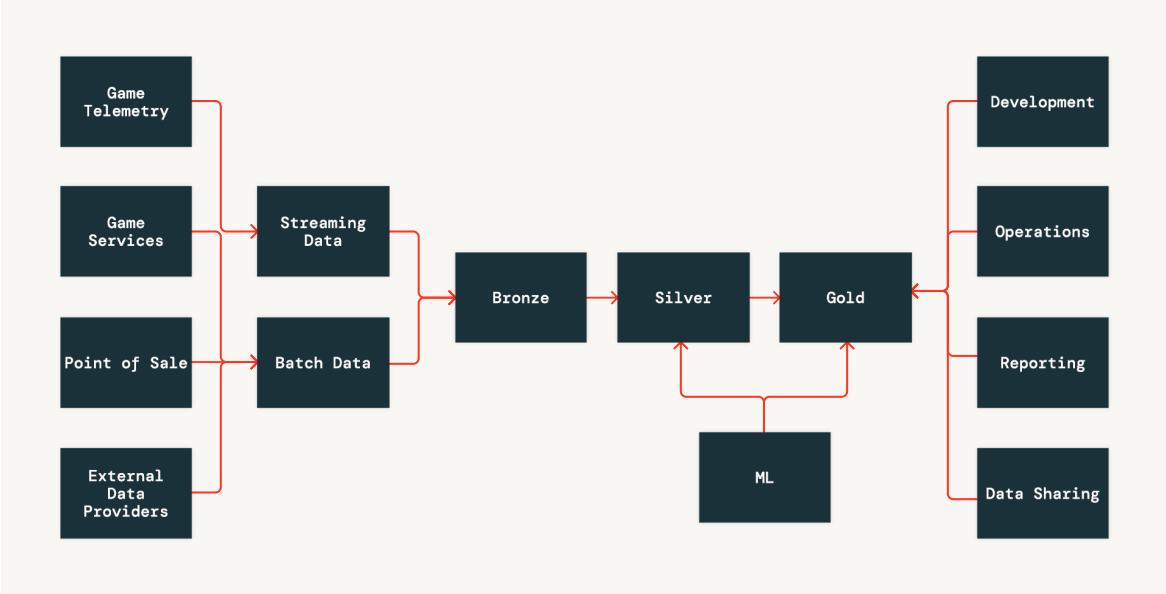
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SHARING DATA ACROSS THE ORGANIZATION

Different studios and game teams will have differentiated dataset requirements. Your data platform must allow for their flexibility, and your strategy should define how they'll share their data or their derived insight across the organization. Determine how you'll:

- Enable teams to collect, store and analyze data from across the organization
- Ensure governance of the data
- Track the data dictionary built for their title
- Get feedback on insight found that might benefit other teams



Step 5: Consider architecture and engineering

As we continue evolving toward the concept of data products within games organizations, we're starting to leverage best practices found in software engineering.

DATA PRODUCTS AND SOFTWARE ENGINEERING

Your environment, like a software engineering project, will evolve over time. Integrating CI/CD-like processes and taking a data product mentality will enable your success as you adapt to market changes. How will you:

- Leverage CI/CD-like processes within your data platform?
- Promote and document changes to your data products?
- Automate and roll back changes as needed?

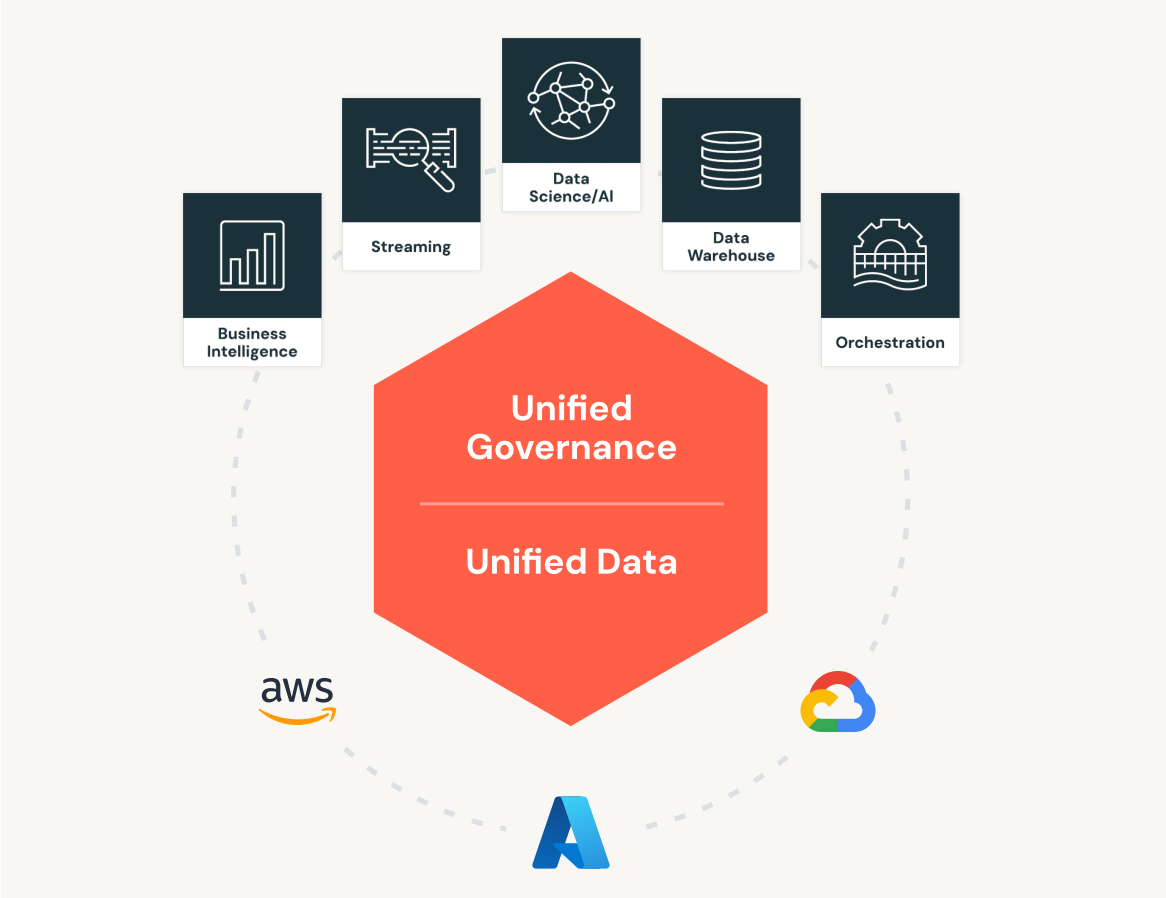
PRIVACY, GOVERNANCE AND AUDITABILITY

Your data architecture needs to be sufficiently hardened to address regulatory requirements, flexible enough to address changes in those requirements and governed to ensure auditability. How will:

- Users discover and get access to new datasets or models?
- Your team audit data access and processes like the right to be forgotten?
- You ensure that data for sensitive groups isn't used in an inappropriate manner?

CONSIDER DATA BIAS EARLY

Try to limit hard-coding heuristics into your data processing. Often these heuristics are based on someone's beliefs and aren't aligned with reality or flexible enough to experiment with and investigate your data. For example, consider a business rule that overemphasizes D1–D3 engagement levels when calculating the pLTV of a player. If the player installed the game on Tuesday, they may not really dig into the title until the weekend and be passed over. Double-checking your assumptions and manual heuristics will help you evolve your player understanding over time.



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Step 6: Data access, analysis and insight

Now the rubber hits the road. Your strategy for how your core and noncore data users will access the organization's data platform will strongly influence the impact data will have. Most of these requirements fall within the domain of data governance. There are many solutions to consider, but Databricks Unity Catalog provides all of these capabilities. Ensuring appropriate data governance encompasses many activities that will involve people across your organization. Creating resources like an internal online portal, confluence pages and/or wiki can help users through these aspects of your platform.

There are many considerations here, including:

- What tooling is available and to whom it's available
- How you'll enable governed data democratization across the organization
- How to make it easy for game developers, product managers, creatives and other noncore data users to generate their own insight
- How your users will find the data they're looking for and understand how to use it
- How to instill trust in your data

DATA ACCESS

Build out processes that allow end users to access data quickly. These processes need to be relatively simple to avoid shadow data teams. Connect your single sign-on provider to your platform, and create descriptive groups to provide consistent access and simplify the management of data access over time.

DATA ONBOARDING

Create a well-defined training program for new team members, or implement a train-the-trainer program. How will you train team members initially, and how will you continue to train them?

DATA DISCOVERY

Leverage a data catalog, ideally one that is self-documenting like Unity Catalog. This will make it easy for your data users to find interesting datasets.

DATA LINEAGE

Ensure that your system provides provenance for metrics and interim datasets. The critical nature of data lineage is starting to be understood within the industry. By being able to see where the data comes from, you help ensure that data users understand its provenance and can trust it. Getting funding for data lineage tooling and head count is often challenging. Leverage a data governance solution like Unity Catalog that provides this by default to ensure your team's trust in produced insight.

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DATA VISUALIZATION

Leverage a tool that queries your data lakehouse directly, like Sigma. Games especially, benefit from near real-time data. Loading data into your visualization tooling adds unnecessary latency to your system. Imagine your Live Operations team releasing a new event, and there's a catastrophic failure in the game. If it takes your dashboards 30 minutes to refresh the dataset, you've just shed a bunch of players. This architecture will also benefit you as a sponge for inbound data during large spikes of traffic. Ensure your tool:

- Integrates well with your chosen data lakehouse
- Enables your noncore data users to create and curate their own visualizations and data investigation efforts
- Doesn't make the central data team a bottleneck for your data users

MINIMIZE DATA DUPLICATION

The more copies of your data, the harder it'll be to track, govern, trust and make the data useful. Pick a data lakehouse that minimizes data duplication, simplifies administration and keeps your data and your insight in a single place. This aligns your data scientists, business analysts and game teams, and ensures they're using the same data. Every hop between solutions causes increased latency, administrative burden and governance issues. When you work with an integrated data platform you can spend more of your time on insight generation.

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Step 7: Create cross-functional data organizations

The creation of technology guilds isn't novel. These will often start with game development, SRE or software development. It's rare to find technology guilds within the context of data because the core teams tend to be much smaller than other communities of practice. Due to the loosely coupled nature of games companies, it can be difficult to drive thoughtful, cross-functional data efforts.

Set a schedule and follow it. In smaller game studios it may not make sense to build a lot of process and oversight for these groups. Instead, opt for monthly meetings or networking sessions. For medium to large games organizations, consider formalizing this effort so noncore data teams will feel more welcome. This is important because these individuals provide the most value through this process.

Building a community takes time and commitment but even then success isn't guaranteed. In talking with several organizations that have worked on these projects, we've found:

- The organization must view data as highly valuable for this to succeed
- Your core data team needs mental space to invest in this effort. If they're overworked, engagement won't occur.
- Creating time for networking and career development helps these efforts succeed
- Publishers with studios that are combative struggle to build these communities
- Tracking engagement, growth, training and aha moments helps invigorate the community and ensure leadership support

A GRASSROOTS COMMUNITY

These groups come in many forms, but they're best exemplified by a data guild. Most resources you find will talk about guilds in general, such as developer guilds or SRE guilds. The concepts, approaches and suggestions made there apply equally here. Most of the time these efforts are started by a passionate individual bringing people together. They're organic and often take a lot of time to build.

ADJUST FOR SIZE

Consider the size of your studio. If your studio only has 5 to 10 potential data users, you won't need as much organizational rigor as you would if you had 100 data users. With a small group, you may only meet once a month or quarterly. With a larger community, you may break into smaller subgroups for biweekly meetings and plan larger, community-wide meetings quarterly where representatives from each subgroup present to the whole.

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DATA GUILDS

Include a high variety of data skills and personas to gather feedback about your data platform from across the organization. Guilds will share their experience, test new capabilities as they're built and help train their teams. Recruit from noncore data teams like game development, product management, server operations, testing and community management. Including these teams and their high variety of data skills will provide valuable feedback on what's working and what isn't within your data estate.

DATA LEADERSHIP COUNCIL

This council owns and promotes cross-functional data initiatives. It provides your data projects the visibility needed to garner funding and maintain focus on high-impact initiatives. The leadership council will typically be run by the chief data officer or another executive sponsor who's highly reliant on quality data (like a chief revenue officer). Often this is the same passionate individual who began the grassroots effort.

DEFINE GOALS FOR THE GUILD

For the guild to be most effective, it's helpful to define functional goals. Examples include:

- Addressing feedback across the organization
- Helping to define the direction of the data platform through roadmap discussions
- Sharing the value of the data platform with the greater organization

PROVIDE A FEEDBACK LOOP

In addition to the guild, ensure you have a way for any end user to provide feedback. This could be an email alias (e.g., data-feedback@) or a form-based survey. It's important to close the loop with individual users to grow trust within the organization. Send out a regular data platform newsletter that highlights changes in the platform, reviews the roadmap and highlights success stories to continuously build buzz.

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Step 8: Monitor and optimize

The final step in your data strategy is the commitment to and implementation of continuous improvement. Through the usage of aforementioned feedback systems you'll have insight into your user base and their needs, which you can use to improve your data strategy. You'll set up metrics to find areas of improvement for your platform's data quality. Establishing metrics will also help with requests for funding to improve your data platform. Metrics to consider and publish include:

- How do you track usage over time? Which teams are participating? Which aren't?
- How often are data systems being used by individuals?
- How many downstream applications and systems are relying on this data?
- What are the NPS style scores from your users?

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The most successful Live Operations–focused games companies are data informed. The shift from Premium titles to Premium+ requires a new way of thinking. A focused data strategy and supporting data platform will help with this shift.

This effort will take time and iteration but will result in significant business impact, including:

- Higher yields in user acquisition through marketing campaign optimization
- Better allocation of resources to high-yield creative efforts
- Delighted and engaged gamers, leading to higher retention and monetization
- Increased lifespan for your titles

One of our favorite quotes related to the value of data for a Live Operations game comes from a VP of data for a F2P title earning over $1B: "If it takes more than 10 minutes for me to ingest data, derive insight and take action based on that insight, I can measure the amount of money we're losing." This quote is great because it hits on three key considerations for the data platform of a LiveOps–forward games company:

1. Game data is near real time
2. It isn't enough to ingest data or even to find insight, you have to take action on it
3. The impact of non-action or slowed action is a loss in revenue

Another great perspective comes from Mojang Studios: "**Everything for us boils down to making better decisions for our players . . . to feed their analysis right back into Minecraft so that we can immediately enhance the gaming experience.**"
— Francisco Rius, Head of Data and AI, Minecraft



At Databricks we work with games companies of all sizes and around the globe to make their data platform strategy a reality. If you'd like to explore how Databricks can help with your journey and why it's the best platform for execution of your data strategy, contact us or reach out to your Databricks Account Executive. Thank you for letting us be a part of your journey to serve the players of the world.

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Quantitative datasets

Quantitative datasets are machine generated, sometimes as derivations from qualitative or telemetric datasets. They're your bread-and-butter datasets for Live Operations. These are the datasets with which you'll begin your data journey. We've outlined some of the most standard datasets to consider as part of your strategy. You'll notice some overlap across the dataset types, as it can sometimes be hard to bucket them.

PLAYER-SPECIFIC DATASETS

1. **Progression:** Where is your player in their journey with your title? These datasets are typically framed in milestones. Examples include: Installed, D7, First purchase, FTUE complete, Joined a clan.
2. **Entitlements:** A subset of progression, but often considered separately because they're strong indicators. What has your player purchased? What have they unlocked?
3. **Transactional:** What purchases have they made? What price level do they typically purchase? Do they prefer one type of transaction vs. another? How often do they purchase?
4. **Preferences:** Which game modes do they prefer? When do they play? What weapons, skills, etc. do they prefer? What type of character types do they engage with?
5. **Experience:** The catchall for everything else, including session information, customer service engagement, usage of systems in the game, current state, etc.

GAME-SPECIFIC DATASETS

1. **Demographic:** Where are players engaging (e.g., what geographies)?
2. **Engagement:** Much of this involves figuring out the averages for your players to find anomalies and change over time. What times do they play? How long is the average session? How many sessions per day or week? 95th percentile for spend?

Qualitative datasets

These datasets generally involve free text fields. It's important to attempt to limit bias, the halo effect and other logical fallacies when leveraging these datasets. Keep in mind that these forms of feedback tend to evoke extremes from players, both positive and negative. As well, how we think about ourselves isn't always aligned with how we act. For this reason, we generally find quantitative and telemetric datasets to be more reliable.

- **Survey data:** Game studios are often highly reliant on this type of player feedback. Our beliefs regarding what we like and what we actually engage with aren't always aligned. Always compare this feedback to how players engage in your title.
- **Player feedback:** The amount of feedback received from players through reviews, forums and social media can be overwhelming. NLP, LLMs and other techniques can help you filter the noise and find insight to help you improve your title.
- **Customer support requests:** This information is useful to community managers and product teams so they can address immediate issues a player may have and gain insight into where you may have bugs. It's also useful for churn prevention and reengagement.
- **In-game chat:** This data is necessary to help address bots, abuse and toxicity. It may also be useful for churn prevention and segmentation projects.

Telemetrics and logs

Telemetrics and logs are your "diamond in the rough" datasets. These data types are typically verbose, require significant parsing and will quickly grow your object storage bill. As a result, it's common either to sample this data rather than retain it all or to derive tabular datasets and store the raw data in low-tier object storage (just in case you need to reprocess it down the line).

- **Quality of service:** Your operations team will be collecting metrics about your player's experience. Did they suffer due to an outage? Did their match fail as a result of a server crash? Are they suffering from latency? All of these data points are helpful when managing VIP players and for reengagement efforts.
- **Session:** Session data is some of your most valuable telemetric data, but it's also quite challenging to use effectively. Most studios pull metadata about sessions instead of keeping full session details. Assuming you have a way to play back a session, keeping full session details can be helpful when trying to understand a player's experience with your title.
- **Match:** Transcripts of matches in games where matches exist are typically hard to follow, but they're especially useful when trying to identify cheaters and find potentially exploitable game mechanics

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A COUPLE OF REMINDERS WHEN CONSIDERING THESE METRICS

- Wherever a metric indicates "User" (e.g., all players), you can replace "User" with "Customer" (paying players) for another highly useful metric
- Most of these metrics can be considered at different levels within the organization — from a publisher level to a studio to a specific game. Organization-wide metrics should have consistent inputs and calculations, but lower-level metrics can be modified (e.g., cLTV at an organization may only consider actual purchases, but at a studio level it could be a prediction based on individual player metrics and their journey).
- Some aggregated metrics (playtime for a user, for example) are also valuable at the individualized level

Engagement metrics

USER COUNT FOCUSED

- **Daily active users (DAU):** The number of unique users who engage with the content or product within a 24-hour period.
    a. If you're multiplatform, can you identify a single user across both platforms? If not, by how much are you overcounting users? How many of your users only show up for new content (compare DAU over the month/quarter vs. MAU to find blips)?
- **Monthly active users (MAU):** The total number of unique users who engage with the content or product over the course of a month
- **Concurrent users (CCU):** How many users are online at any given time? What's your average? Is that average growing or shrinking and at what rate? How does your CCU look during an event as compared to baseline? Are there patterns during the day?
    a. Example question: What day of the week and time of day has the highest CCC (Customer)?
    b. Usage: If a pattern emerges, this could be a good target release day/time for new events. Target for customers to try to maximize spend.
- **Stickiness (DAU/MAU ratio):** A measure of user engagement and retention, calculated by dividing DAU by MAU

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- **Retention rate:** The percentage of users who continue to use the product or service over a specific time period after their initial engagement.
    a. If you're seeing especially bad D1 and D3 numbers, you aren't hooking your players, they're not getting the game or your FTUE (first-time user experience) has an issue. If you're seeing a big drop-off between D60 and D90, or D90 and D180, you likely have an issue with elder gameplay.
- **Churn rate:** The opposite of your retention rate (e.g., 1 - RR). This number is used when trying to frame the cost of doing nothing or the lost revenue potential through churn.
    a. Often expressed as D#. For example, D1 retention, day one. D7, D30, D90 and D365 are the most standard expressions of this metric.
    b. If the churn rate for customers is 5% over a 30-day period, your cLTV = $35 and your MAC is 1,000,000, you have $1.75M walking out the door each month, and a 10% improvement is worth $175K. The numbers are more complex than this, but you get the point.

SESSION FOCUSED

- **Session length:** The amount of time a user spends interacting with the product or content in a single session
- **Session interval:** The average time between a user's sessions
- **Session frequency:** How often does the user engage with the product? Sessions are divided by day.

PLAYER JOURNEY FOCUSED

- **Average playtime per user:** The total amount of time a user spends playing a game or interacting with an app
- **Player progression speed:** The rate at which a player advances through the levels or stages of a game

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- **Milestone achievement:** These metrics track completion of a milestone, how long it took to get to that milestone and related metadata. You may track a milestone against: Achieved, Playtime to reach, # of sessions to complete and days to complete. Examples include: tutorial completion, joining a clan, PVP engagement, engagement with social features, first purchase and/or rewarded ad opt in. A few common examples include:

    a. **First-time user experience (FTUE):** Where did the player make it during their first session, on their first day? First impressions are critical. Tracking progress made during the FTUE will enable you to improve how your players perceive you and your D1+ metrics.

    b. **Tutorial completion rate:** The percentage of new users who complete the introductory tutorial or guide

    c. **Level completion rate:** The percentage of players who complete a specific level or stage in a game

    d. **Achievement completion rate:** The proportion of users who complete set achievements or milestones within the game or app

PLAYER EXPERIENCE FOCUSED

- **Experience ratios:** The most common of these are W/L (Win/Loss) and K/D (Kill/Death), but these only apply to specific gametypes. A match 3 game might look at Moves/Win, attempts to pass or time to win. These metrics help you determine player skill and potential frustration and are helpful with personalization efforts.
- **Social engagement:** The number of times content is shared by users on social media platforms
- **Community engagement:** The level of interaction and participation of users within a community such as forums or social media groups
- **Player feedback and ratings:** The qualitative and quantitative feedback provided by players, often in the form of ratings or reviews
- **Net promoter score (NPS)**: A metric that measures customer loyalty and satisfaction based on their likelihood to recommend the product or service to others

Monetization metrics

ADVERTISING AND USER ACQUISITION (UA) FOCUSED

- **Customer acquisition cost (CAC):** The average expense incurred to acquire a new customer, including marketing and sales costs
- **Cost per install (CPI):** Similar or even the same as your CAC, but focused on UA-specific costs
- **Click-through rate (CTR):** The percentage of people who clicked on your ad creative. This is most useful to see if your creatives are getting your players interested and which ad creatives are most effective.
- **Cost per click (CPC):** The expense accrued for each click made by users in a pay-per-click (PPC) advertising campaign
- **Return on advertising spend (ROAS):** The amount of revenue generated per unit of advertising spend. This is used to gauge the effectiveness of ad campaigns.
- **Payback period:** The time required to recover the investment in acquiring a customer or launching a marketing campaign
- **Install source:** Identifies which ad provider brought this player to your game. **Note:** This only tells you the end of their journey, but it may have taken several ads from different sources before they clicked through. Research multi-touch attribution models to understand the journey aspect of UA more deeply.
- **Effective cost per mille (eCPM):** A high eCPM generally means that you're paying more than the average for your ad impressions. **Note:** eCPM is calculated differently by the ad networks and can be hard to compare. Many things go into this number.

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REVENUE FOCUSED

- **Lifetime value (LTV, cLTV and pLTV):** The dollar value attributed to your player.
  a. This comes in many forms:
     i. pLTV (player lifetime value): Total revenue attributed to a player. A current statement.
     ii. pLTV (predicted lifetime value): Total projected revenue over the full lifetime of a player or customer. A long-term statement.
     iii. cLTV (customer lifetime value): One of these first two metrics, but only considering players who have monetized in your title
  b. Key considerations:
     i. How do you normalize across long-term players vs. new players? At what point do you feel confident enough to predict their LTV?
     ii. Example question: What % of customers make up 80% of my revenue? What is the lowest LTV for that group?
     iii. Usage: This could help you define your VIP program cutoff points
- **Average transaction value (ATV):** Total transactional revenue divided by number of transactions. This should be considered separately for each type of transaction (e.g., premium purchase price, IAP, battle pass, rewarded ad). **Note:** This is an example of a metric that's useful at the global level, the cohort and the individual. Many of these metrics can be used to help understand the health of a title, but they can also be helpful at the individual level as part of a personalization strategy.
- **Time to purchase (TTP):** How long does it take, on average, for a player to make their first purchase and convert to a customer?
- **Average revenue per user (ARPU):** Total revenue generated over a period of time divided by that length of time's unit of measure then divided by the number of players. Critically important when considering the cost of acquisition, retention and operations.
- **Average revenue per paying user (ARPPU):** Total revenue generated over a period of time divided by that length of time's unit of measure then divided by the number of paying customers
- **Average revenue per daily active user (ARPDAU):** Global KPIs like ARPU can get skewed over time, and because they're based on such large numbers, they don't move much over time. Looking at numbers that describe a shorter period of time enables you to understand the here and now.
  a. Example question: What was my ARPDAU over the last 30 days? How does it compare to my overarching ARPU? What does the trend look like?

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Is your game data making you smarter?

Databricks is the end-to-end data and AI platform that supports everything from creating a 360-degree view of your player base to building ML models that serve more immersive experiences. From indie to AAAA, discover why studios of all sizes run on Databricks.

Learn more about Databricks for games

About Databricks

Databricks is the data and AI company. More than 10,000 organizations worldwide — including Block, Comcast, Condé Nast, Rivian, Shell and over 60% of the Fortune 500 — rely on the Databricks Data Intelligence Platform to take control of their data and put it to work with AI. Databricks is headquartered in San Francisco, with offices around the globe, and was founded by the original creators of Lakehouse, Apache Spark™, Delta Lake and MLflow. To learn more, follow Databricks on LinkedIn, X and Facebook.

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